

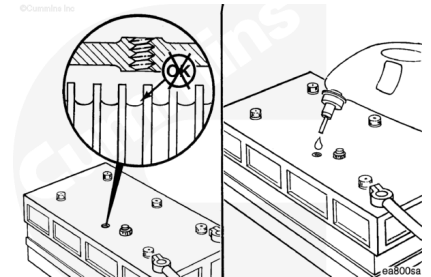
Trainings ▾

Initial Check

If conventional batteries (**not** maintenance-free batteries) are used, check the electrolyte level in each cell. If it is low, add distilled water to bring the level up to the top of all plates. Refer to the battery manufacturer's specifications.

Note : If water is added to the battery, the battery must be charged before any testing can be accomplished.

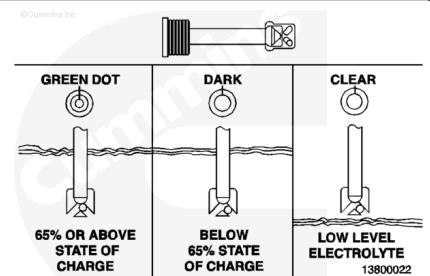
With conventional batteries, check the electrolyte specific gravity with a hydrometer. To check the specific gravity, the battery temperature will be 26.67°C [80°F]. All cells will **not** be less than 1.230. The difference between high and low readings will **not** exceed 0.050. If the readings indicate a difference of more than 0.050, replace the battery. If the readings indicate less than 0.050 difference but one or more cells indicate less than 1.230, recharge the battery.



Battery Specific Gravity at 26.67°C [80°F]

Specific Gravity	State of Charge
1.260 to 1.280	100 percent
1.230 to 1.250	75 percent
1.200 to 1.220	50 percent
1.170 to 1.190	25 percent
1.100 to 1.130	Discharged

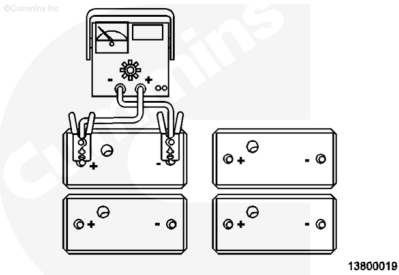
Check the “eye” on the maintenance-free battery. Refer to the manufacturer's specifications.



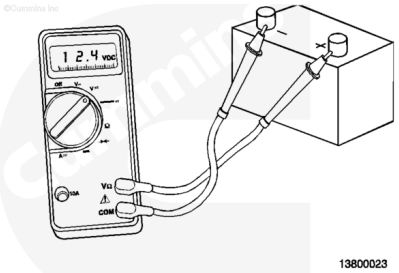
Note : For conventional batteries, remove the vent caps prior to removing surface charge. If a blue haze is observed in any cell while removing the surface charge,

replace the battery. After the surface charge has been removed, install all vent caps.

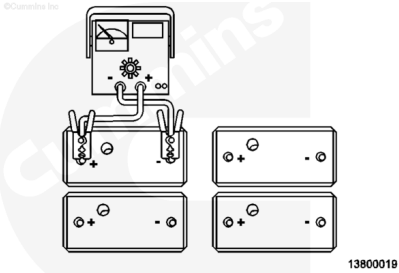
Remove the surface charge by attaching the battery to a 300-amp load for 15 seconds for heavy-duty batteries.



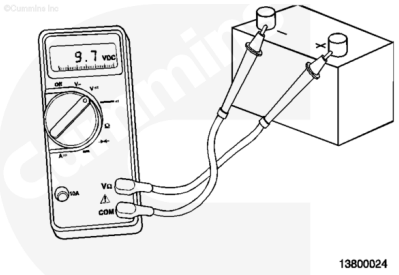
Remove the load and wait 1 minute; if the battery voltage is greater than or equal to 12.4 VDC, continue testing. If the voltage is below 12.4 VDC, recharge or replace the battery.



Load-test the batteries at 1/2 of the cold cranking amp rating of the battery (rating at -18°C [0°F]) for 15 seconds.



Check the battery voltage and compare to the table below:



Temperature and Voltage Relationship								
Temperature (F)	70	60	50	40	30	20	10	0
Temperature (C)	21	16	10	4	-1	-7	-12	-18
Minimum Voltage (VDC)	9.6	9.5	9.4	9.3	9.1	8.9	8.7	8.5

Turn the load off.

Replace the battery if it does **not** meet the above specifications.

Last Modified: 13-Mar-2002